

Name Key
(print)

Total = 100 points

Please show all your work.

1. (20 pts) For the chemistry test scores the mean is 75 with a standard deviation is 6.

For the physics test scores the mean is 78 with a standard deviation of 5.

a) (10 pts) Suppose Brian took both exams and he got a 70 on the chemistry exam and a 72 on the physics exam. Use the z-score to decide on which test did Brian perform better.

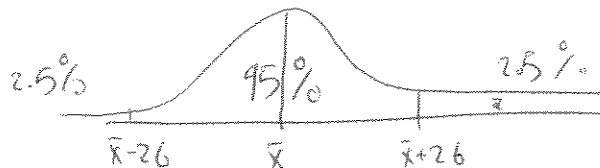
$$Z_{\text{chem}} = \frac{70 - 75}{6} = -0.833$$

$$Z_{\text{phys}} = \frac{72 - 78}{5} = -1.2$$

Brian did better on chemistry exam.

b) (10 pts) What percentile does a score of 87 on the chemistry test correspond to? (Assume a bell shaped distribution; begin by computing the z-score.)

$$Z = \frac{87 - 75}{6} = 2$$

Percentile 97.5 (^{98 or} 97 also OK)

2. (12 pts) A card is selected at random from a standard deck of cards. Find the probability that:

a) (6 pts) a randomly selected card is a black suit or a king

$$P(\text{black or K}) = \frac{26}{52} + \frac{4}{52} - \frac{2}{52} = 0.538$$

Answer 0.538 or 53.8%

b) (6 pts) a randomly selected card is a five or a ten

$$P(5 \text{ or } 10) = \frac{4}{52} + \frac{4}{52} = 0.154$$

Answer 0.154 or 15.4%

3. (12 pts) A 6-person jury must be selected at random from a pool of 20 people that has 12 men and 8 women. Find the probability of selecting a jury that consists of 4 men and 2 women.

$$\frac{{}^{12}C_4 \times {}^8C_2}{{}^{20}C_6} = \frac{495 \times 28}{38760} = .358$$

Answer .358 or 35.8%

4. (20 pts)

		Gender		
		Men	Women	Total
Dominant Hand	Left	75	60	135
	Right	465	365	830
	Total	540	425	965

A person is selected at random from the sample. Find the following probabilities:

a) (5 pts) the person is right-handed man.

$$\frac{465}{965} = .482$$

Answer .482 or 48.2%

b) (5 pts) the person is left-handed or a woman.

$$\frac{75 + 60 + 365}{965} = .518$$

Answer .518 or 51.8%

c) (5 pts) Given that the person is a male, what is the probability that he is left-handed?

$$P(\text{Left} | \text{Male}) = \frac{75}{540} = .139$$

Answer .139 or 13.9%

d) (5 pts) If we select 3 people from the sample (without replacement), what is the probability that at least 1 is a right-handed woman?

$$P(\text{at least 1 R-H-Women}) = 1 - P(\text{no R-H-Women})$$

$$= 1 - \frac{{}^{465}C_3}{{}^{965}C_3} = .76$$

$$1 - \frac{600}{965} \times \frac{599}{964} \times \frac{598}{963} = .76$$

Answer .76 or 76%

5. (8 pts) The board of directors for a company has 15 members. One member is the president, another is the vice president and another is the treasurer. In how many ways can these positions be assigned?

$$15P_3 = 2730$$

Answer 2730

6. (20 pts) The probability that a person in the U.S. has A+ blood type is 34%.

a) (5 pts) If a person in the United States is selected at random, find the probability that she/he does not have A+ blood type.

$$1 - .34 = .66$$

Answer .66 or 66%

b) (7 pts) If five unrelated people in the U.S. are selected at random, find the probability that all five have A+ blood type.

$$(.34)^5 = .00454$$

Answer .00454 or .454%

c) (8 pts) If five unrelated people in the U.S. are selected at random, find the probability that at least one of the five has A+ blood type.

$$P(\text{at least one has } A^+) = 1 - P(\text{No one has } A^+) \\ = 1 - .66^5 = .875$$

Answer .875 or 87.5%

7. (8 pts) Find the number of distinguishable ways (permutations) the letters in the word CLASSEXAM can be written.



$$\frac{9!}{2! 2!} = 90720$$

Answer 90720